

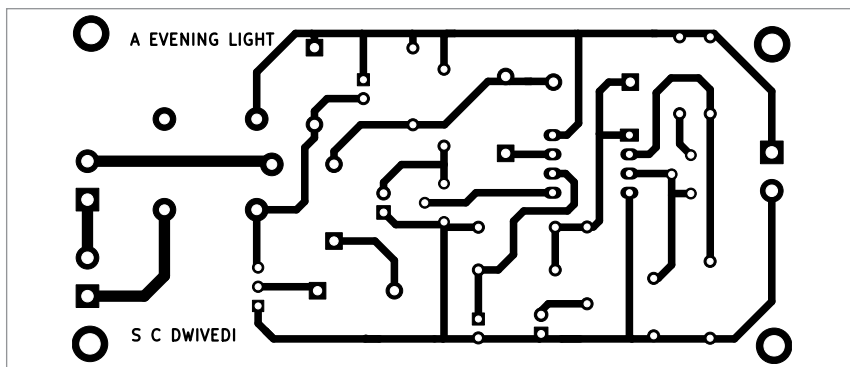


Fig. 3: Actual-size PCB

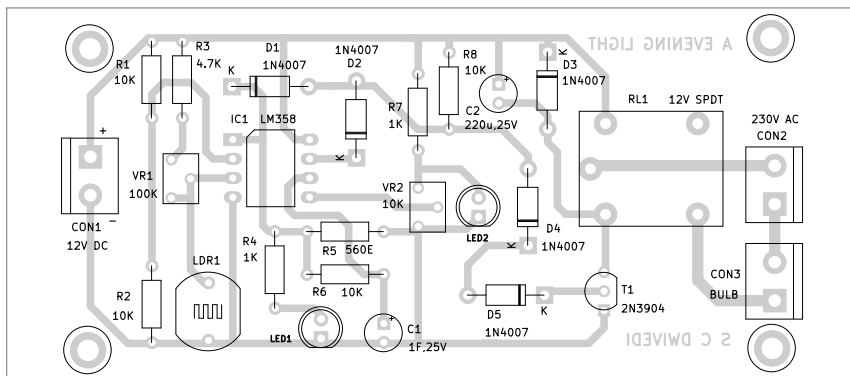


Fig. 4: Components side

now begins discharging through R6 and R5. By the evening, C1 is fully discharged, and the circuit is ready again for the next operating cycle. R5 is essential to ensure that C1 discharges even when the power supply is on. The device does not need to be reset in the daytime and will continue to switch on the light every evening for a few hours without human intervention in a locked house.

Experimentally, it has been observed that the output of the comparators, when low, is not zero volts. A small voltage—about 0.8 volts—exists at the output of the comparators when the output is low. Due to diodes D1 and D2, which form part of the AND gate, the voltage at the junction of the anodes of D1, D2, and D4 is around 1.3 to 1.4 volts, even when the output of either comparator is low. This is sufficient to turn on transistor T1. Hence, D4 and D5 are connected in the base circuit to

absorb this voltage and ensure that the transistor conducts only when the output of both comparators is high.

## Construction and testing

An actual-size, single-sided PCB layout for the automatic evening light is shown in Fig. 3, with its component layout in Fig. 4. After assembling the circuit on the PCB, enclose it in a suitable box, positioning all the switches and LEDs on the top of the box. Fix the LDR on the top of the box. Mount CON1 on the front side of the box. Mount CON2 and CON3 on the back side of the box. The device can also be assembled on a general-purpose board or a Veroboard.

Keep the wiper of VR2 towards the anode of LED2 so that about 2 volts are applied to the non-inverting input of IC1B. In the evening, adjust VR1 so that LED1 and the lamp turn

## PARTS LIST

### Semiconductors:

|                  |                          |
|------------------|--------------------------|
| IC1 (IC1A, IC1B) | - LM358 dual op-amp      |
| T1               | - 2N3904 NPN transistor  |
| D1-D5            | - 1N4007 rectifier diode |
| LED1, LED2       | - 5mm LED                |

### Resistors (all 1/4-watt, $\pm 5\%$ carbon):

|                |                 |
|----------------|-----------------|
| R1, R2, R6, R8 | - 10k $\Omega$  |
| R3             | - 4.7k $\Omega$ |
| R4, R7         | - 1k $\Omega$   |
| R5             | - 560 $\Omega$  |

### Capacitors:

|    |                                 |
|----|---------------------------------|
| C1 | - 1F, 2.7V super capacitor      |
| C2 | - 220 $\mu$ F, 25V electrolytic |

### Miscellaneous:

|           |                                  |
|-----------|----------------------------------|
| CON1-CON3 | - 2-pin connector                |
| BATT.1    | - 12V battery/12V adaptor        |
| RL1       | - 12V, SPST relay                |
| LDR1      | - Small light-dependent resistor |
|           | - 60-watt bulb for lamp          |

on. After a few hours, adjust VR2 so that the light switches off.

## Cut-off time calibration

The output of comparator IC1A will be high in darkness. The output of comparator IC1B will remain high as long as the voltage at C1 is lower than the voltage at the wiper of VR2, which is connected to pin 5 of IC1B. The supercapacitor C1 will charge to 63.2% of the LED1 voltage (2V), which is approximately 1.264V, after 10,000 seconds (or 2 hours and 47 minutes), and to 86.5% of the LED1 voltage (approximately 1.73V) after 20,000 seconds (or 5 hours and 33 minutes).

Therefore, the duration for which the lamp remains on can be set by adjusting the voltage at the wiper of VR2. For example, to switch the light off 2 hours and 47 minutes after it turns on in the evening, set the voltage at pin 5 of IC1B to 1.264V using VR2. This enables the calibration of the switch-off time between 2 to 6 hours using VR2. **EFY**

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